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Regional Octagonal Tile Rewriting Grammars

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Abstract: Octagonal Tile Rewriting Grammars are modern class of octagonal tiling based Array Rewriting Grammar models on picture processing. They form the octagonal variant of triangular (hexagonal) Tile rewriting Grammars. We introduce Regional Octagonal Tile Rewriting Grammar (ROTRG) and generate images. We have compared this model with Context Free Octagonal Array Grammar and Octagonal Tile Rewriting Grammar.

Keywords: Picture language, Tiling system, Picture grammar, Octagonal arrays, Rewriting Grammars

1. Introduction

Picture languages generated by grammars [3,10,11,12,13] have been advocated since seventies for problems arising in the framework of pattern recognition and image analysis. Regional tile rewriting grammars (RTRG) and its hexagonal variant (RHTRG) [6] are recently introduced tiling based array rewriting grammar models on picture processing. These models are resulting from the restriction of rectangular (hexagonal) Tile Rewriting Grammars [1, 5, 14]. RTRG (RHTRG) has greater generative capacity than that of Siromoney's (hexagonal) Array grammar [2, 3, 6] and context free (hexagonal) Prussa Grammar [9].

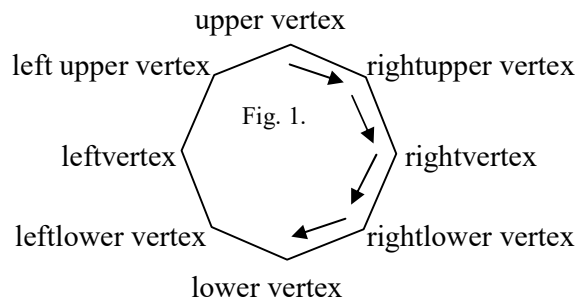
Triangular Tile Rewriting Grammars (TTRG) [7] and its regional version (RTTRG) introduced in [4], combined the features of Triangular Tiling Systems (TTS) and Triangular Local Picture Languages (TLOC). A typical rule of RTTRG grammar has a left and right part, of all triangular pictures of unspecified but equal size (isometric). In [4], we compared the class RTTRG with Context Free Triangular Array Grammars (CFTAG) and TTS. It has been proved that RTTRG has more generative capacity than CFTAG and also showed that the family of languages generated by RTTRGs is included in the family of languages generated by TTRGs. Recently, we introduced Octagonal Tile Rewriting Grammars (OTRG)[8] augmented with the features of Octagonal Tiling Systems (OTS) and Octagonal Local Picture Languages (OLOC).

Motivated by the above studies, we introduce Regional Octagonal Tile Rewriting Grammars (ROTRG). We also study some closure properties of it. It is compared with other models.

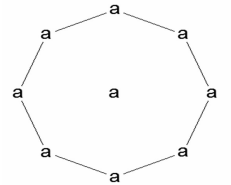
II. Preliminaries

Definition 2.1:

Let Σ be a finite alphabet of symbols. An octagonal picture p over Σ is an octagonal array of symbol of Σ_0 . We consider octagons of the type:



For example, an octagonal picture over the alphabet $\{a\}$ is



Definition 2.2:

The set of all octagonal arrays over of the alphabet Σ is denoted by Σ_0^{**} . An octagonal picture language L over Σ is a subset of Σ_0^{**} .

Definition 2.3:

For $p \in \Sigma_0^{**}$, let $\hat{p}$ be the octagonal array obtained by surrounding p with a special boundary symbol $\# \notin \Sigma$, for example

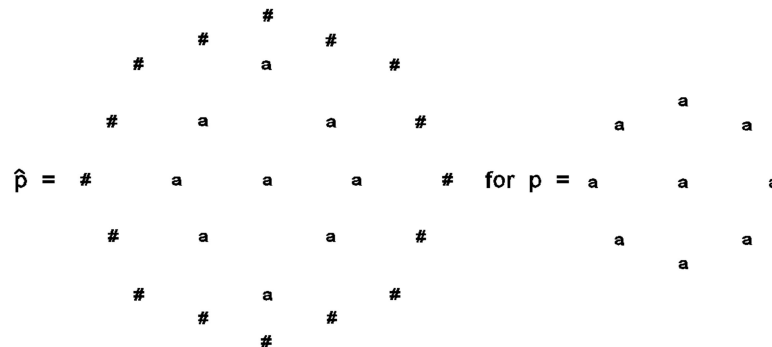


Fig. 2. Octagonal Array

Definition 2.4:

With respect to axes x_1, x_2, x_3, x_4 , the co-ordinates of each element of octagonal picture can be fixed.



Given a picture $p \in \Sigma_0^{**}$, let 'a' denote the number of elements in the border p from upper vertex to right upper vertex in the direction, 'b' denote the number of elements in the border of p from right upper vertex to right vertex in the direction, 'c' denote the number of elements in the border of p from right vertex to right lower vertex in the direction, 'd' denote the number of elements in the border of p from right lower vertex to lower vertex in the direction. The directions are fixed with origin of reference as the upper vertex, having coordinates $(1, 1, 1, 1)$.

Definition 2.5:

The four tuple (a, b, c, d) is called the size of the picture p denoted by $|p| = (a, b, c, d)$. Let p_{ijkl} denote the symbol in p with co-ordinates (i, j, k, ℓ) where $1 \leq i \leq a, 1 \leq j \leq b, 1 \leq k \leq c, 1 \leq \ell \leq d$. Let $\Sigma_0^{(a,b,c,d)}$ be the set of octagonal pictures of size (a, b, c, d) .

Definition 2.6:

Given an octagonal picture p of size (a, b, c, d) , we denote by $B_{e,f,g,h}(p)$ the set of all octagonal subpictures of p of size (e, f, g, h) , where $e \leq a, f \leq b, g \leq c, h \leq d$. Each member of $B_{2,2,2,2}(p)$ is called an octagonal tile.

Definition 2.7:

A non-convex octagon ABCDEFGH as shown in Fig. 1 is called an arrowhead if $|BC| = |HA|$, $|CD| = |GH|$, $|DE| = |FG|$, BC is parallel to AH and CD is parallel to GH and DE is parallel to FG. It is noted that the opposite sides $|AB|$ and $|EF|$ are equal and parallel. $|AB|$ is the thickness of the arrowhead. BCDE is the outermost edge and FGHA is the innermost edge.

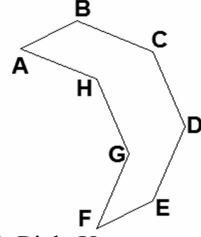


Fig. 3. Right Upper arrowhead

Depending on the direction of a vertex an arrowhead is classified as upper arrowhead (U), right upper arrowhead (RU), right arrowhead (R), right lower arrowhead (RL), lower arrowhead (L), left lower arrowhead (LL), left arrowhead (L), left upper arrowhead (LU).

Definition 2.8:

If PQRSTUWV (Fig. 2.) is an octagon and ABCDEFGH is an upper right arrowhead, then the arrowhead can be catenated to the octagon in the right upper direction .

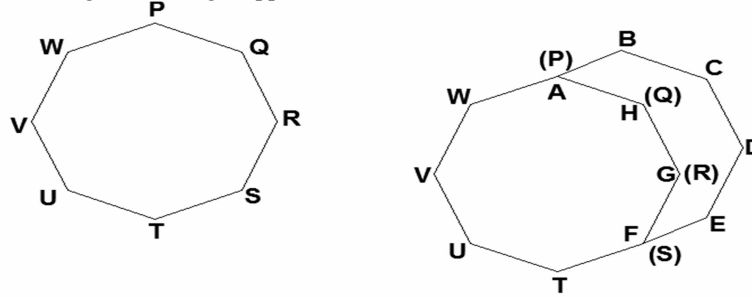


Fig. 4. Arrowhead Catenate to the Octagon in the Right Upper Direction

It is noted that $|WB| = |WA| + |AB| = |WP| + |PB|$ and $|TE| = |TF| + |FE| = |TS| + |SE|$.

Similarly the remaining arrowhead catenations can be defined.

Definition 2.9:

Let Γ and Σ be two finite alphabets and $\pi : \Gamma \rightarrow \Sigma$ be a mapping which we call, a projection. Let $p \in \Gamma_0^{**}$ be octagonal picture. The projection by mapping π of p is the picture $p' \in \Sigma_0^{**}$ such that $p'(i, j, k, \ell) = \pi(p(i, j, k, \ell))$ for all $1 \leq i \leq a, 1 \leq j \leq b, 1 \leq k \leq c$ and $1 \leq \ell \leq d$, where (a, b, c, d) is the size of the octagonal picture.

Definition 2.10:

Let $L \subseteq \Gamma_0^{**}$ be an octagonal picture language. The projection by mapping π of L is the language $L' = \{p' / p' = \pi(p), \forall p \in L\} \subseteq \Sigma_0^{**}$.

Definition 2.11:

Let Γ be a finite alphabet. An octagonal picture language $L \subseteq \Gamma_0^{**}$ is called local if there exists a finite set Δ of octagonal tiles over $\Gamma \cup \{\#\}$ such that $L = \{p \in \Gamma_0^{**} / B_{2,2,2,2}(\hat{p}) \subseteq \Delta\}$. The family of octagonal local picture languages will be denoted by $L(OLOC)$.

Example:1

Let $\Sigma = \{1, 2, 3\}$ be a finite alphabet.

$$\Delta = \left\{ \begin{array}{cccccccccccccccc} & & \# & & & \# & & & \# & & & \# & & & & \\ \# & & \# & & 1 & & \# & & 2 & & \# & & & & & \\ \# & 1 & & \# , 2 & 1 & & \# , 1 & 1 & & \# & & & & & & \\ 2 & & 2 & & 1 & & 1 & & 2 & & \# & & & & & \\ & 1 & & & & 2 & & & & & & & & & & \\ & 2 & & & & 1 & & & 2 & & & & & \# & & \\ 1 & & 1 & & 2 & & 2 & & 1 & & 2 & & 1 & \# & \# & 2 \\ 2 & 2 & & \# , \# & 1 & & \# , \# & 2 & & 2 , \# & 1 & & 1 , & & & \\ 1 & & \# & & \# & & \# & & \# & & 1 & & \# & & & 2 \\ & \# & & & & \# & & & \# & & \# & & & & & \\ & & \# & & & & & 1 & & & & & & & & \\ & \# & & 1 & & 2 & & & 2 & & & & & & & \\ \# & & 2 & & 2 , 1 & & 1 & & 1 & & & & & & & \\ 1 & & & 1 & & 2 & & & & 1 & & & & & & \\ & & 2 & & & & & 1 & & & & & & & & \end{array} \right\}$$

Then

$$L(\Delta) = \left\{ \begin{array}{cccccccccccccccc} & & & & & & 1 & & & & & & & & & \\ & & & & & & 2 & & & & 2 & & & & & \\ & & & & & 1 & & 1 & & & 1 & & & & & \\ 2 & & 1 & & 2 & & & 2 & & 2 & & 2 & & 2 & & \\ 1 & 1 & & 1 & , & 1 & & 1 & & 1 & & 1 & & 1 & , & \dots \\ & & & & & & & & & & & & & & & \\ 2 & & 2 & & & 2 & & 2 & & & 2 & & 2 & & & \\ & & 1 & & & & & & & & & & & & & \\ & & & & & & & & & 1 & & & & & & \\ & & & & & & & & & 2 & & & 2 & & & \\ & & & & & & & & & & & & & & & 1 \end{array} \right\}$$

$L(\Delta)$ is the Octagonal Local Picture Languages using the above octagonal tiles.

Definition 2.12:

Let Σ be a finite alphabet. An octagonal picture language $L \subseteq \Sigma_0^{**}$ is called recognizable if there exists an octagonal local picture language L' (given by a set of octagonal tiles) over an alphabet Γ and a mapping $\pi : \Gamma \rightarrow \Sigma$ such that $L = \pi(L')$.

The family of all recognizable octagonal picture languages will be denoted by $L(\text{OREC})$.

In example (1) replace 'a' in the place of 1 and 2, we will get recognizable octagonal picture languages, that is $\pi(1) = \pi(2) = a$.

Definition 2.13:

An octagonal tiling system (OTS) T is a 4-tuple $(\Sigma, \Gamma, \pi, \theta)$, where Σ and Γ are two finite sets of symbols, $\pi : \Gamma \rightarrow \Sigma$ is a projection and θ is a set of octagonal tiles over the alphabet $\Gamma \cup \{\#\}$.

Definition 2.14:

The octagonal picture language $L \subseteq \Sigma_0^{**}$ is tiling recognizable if there exists a tiling system $T = (\Sigma, \Gamma, \pi, \theta)$ such that $L = \pi(L(\theta))$.

Definition 2.15:

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An octagonal array grammar $G = (N, I, T, P, S, L)$ where N, I and T are finite sets of non-terminals, intermediates and terminals respectively. P is a finite set of productions, $P = P_1 \cup P_2$ and $S \in N$ is the start symbol. For each A in I , L_A is an intermediate language which is a regular CF and CS string language written in the appropriate arrowhead form. An arrowhead is written in the form $\{...<v>...\}$ where $<v>$ denotes the vertex and the arrowhead is written in the clockwise direction $L = \{L_A / A \in I\}$.

P_1 consists of an initial rule of one of the following form

- | | | | |
|-----------------------|--|-----------------------|--|
| (1) $S \rightarrow O$ | | (2) $S \rightarrow O$ | |
| (3) $S \rightarrow O$ | | (4) $S \rightarrow O$ | |
| (5) $S \rightarrow O$ | | (6) $S \rightarrow O$ | |
| (7) $S \rightarrow O$ | | (8) $S \rightarrow O$ | |

where $S' \in N$, $S' \neq S$ and O is an octagonal array over T .

G is regular if the rules of P_2 are of the forms

- | | | | |
|-------------------------|--|-------------------------|--|
| (1) $S_1 \rightarrow A$ | | (2) $S_1 \rightarrow A$ | |
| (3) $S_1 \rightarrow A$ | | (4) $S_1 \rightarrow A$ | |
| (5) $S_1 \rightarrow A$ | | (6) $S_1 \rightarrow A$ | |
| (7) $S_1 \rightarrow A$ | | (8) $S_1 \rightarrow A$ | |

where $S_1, S_2 \in N$, $S_1, S_2 \neq S$ and $A \in I$.

Furthermore, if an initial rule is P_1 is of the form (r) , $r = 1, 2, \dots$ then P_2 does not contain any rule of the form $(r+1)$ if r is odd, $(r-1)$ if r is even. Also P_1 and P_2 do not contain rules of both the form (r) and $(r+1)$, $r = 1, 3, 5, 7$.

G is CF if the rules are of the form $S_1 \rightarrow \alpha_{i1} \dots \alpha_{ik} (k \geq 1)$ where $S_1 \in N$; $S_1 \neq S$ and $\alpha_i \in (N - \{S\}) \cup I$ ($1 \leq i \leq k$) and α_j denotes any one of the six arrowhead concatenations ($1 \leq j \leq k-1$).

In particular, G is called $(X : R)$ OAG or $(X : CF)$ OAG or $(X : CS)$ OAG, for $X \in \{R, CF, CS\}$ according as all the intermediate languages are regular or at least one of them is CF or at least one of them is CS.

Derivations proceed as follows:

First, an initial rule of P_1 is applied and then the rules of P_2 are applied sequentially as in string grammars until all the non-terminals are replaced, resulting in a string of the form

$O \alpha_{i1} \alpha_{i2} \dots \alpha_{in}$ where $A_i \in I$ ($1 \leq i \leq n$).

In the second phase of derivations, A_1 replaced by an arrowhead from L_{A_1} and catenated to an octagonal array O according to the arrowhead catenation symbol between O and A_1 . This is continued until A_n is replaced, resulting in an octagonal array of terminals. The length of the arrowhead is determined by the condition for arrowhead catenation.

III. Regional Octagonal Tile Rewriting Grammar

In this section, we introduce a grammar with a specified set of tiling named regional. The definitions given below are analogous for the case of Octagonal pictures. The picture p given below refers to an octagon array.

Definition 3.1:

A pixel is an element of an octagonal picture $q(i, j, k, \ell)$. If all pixels are identical to $x \in \Sigma$, the octagonal picture is called x -homogenous or x -picture. The domain of a picture q of size (a, b, c, d) is the set of $\text{dom}(q)$ consists of all positions of pixels in q . A subdomain $\text{dom}_s(q)$ of $\text{dom}(q)$ is a set of positions of pixels in q corresponding to an octagonal subpicture s of size (i, j, k, ℓ) if $1 \leq i \leq a, 1 \leq j \leq b, 1 \leq k \leq c, 1 \leq \ell \leq d$. A subdomain is called x -homogeneous when its associated octagonal subpicture is a x -picture. x is called the label of the subdomain.

Definition 3.2:

A homogeneous partition of an octagonal picture q is any partition $Q = \{\text{doms}_1, \text{doms}_2, \text{doms}_3, \text{doms}_n\}$ of $\text{dom}(q)$ into homogeneous subdomains $\text{doms}_1, \text{doms}_2, \dots, \text{doms}_n$. The unit partition of q , written $\text{unit}(q)$, is the homogeneous partitions of $\text{dom}(q)$ defined by single pixels. A homogeneous partition is called strong if adjacent subdomains have different labels. The unique partitions given by strong homogeneous partition of an octagonal picture q is $\pi^*(q)$.

Definition 3.3:

An Octagonal Tile Rewriting Grammar is a 4-tuple (N, S, R, Σ) , where Σ is the terminal alphabet, N is a set of non-terminal symbol. $s \in N$ is the starting symbol. R is a set of rules of two kinds of forms:

Fixed size : $X \rightarrow s$,

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where $X \in N, s \in \Sigma$.

Variable size : $X \rightarrow \omega$,

where $X \in N, \omega$ is set of octagonal tiles over $N \cup \{\#\}$ such that local octagonal language $L=L(\omega)$ admits a strong homogeneous partition for each $q \in L$.

Definition 3.4:

A homogeneous partition is regional (RO) if and only if distinct subdomains have distinct labels. An octagonal picture q is regional if it admits a RO partition. An octagonal language is regional if all its pictures are regional.

Definition 3.5:

A Regional Octagonal Tile Rewriting Grammar (ROTRG) is a 4-tuple (N, S, R, Σ) , where Σ is the terminal alphabet, N is the set of non-terminal symbols. $S \in N$ is the starting symbol, R is the set of rules of the form:

Type (1) Fixed size : $X \rightarrow s$, where $X \in N, s \in \Sigma$

Type (2) Variable size : $X \rightarrow \omega$, where $X \in N, \omega$ is a set of octagonal tiles over $N \cup \{\#\}$ such that local octagonal language $L=L(\omega)$ admits a strong homogeneous partition for each $q \in L$.

Picture derivation is defined as follows:

Consider an regional octagonal tile rewriting grammar $G = (N, S, R, \Sigma)$. Let $q, q' \in (\Sigma \cup N)^{(a, b, c, d)}$ be pictures of identical size. Let π, π' be the homogeneous partitions of $\text{dom}(q)$ with $\pi = \{\text{dom}_1, \text{dom}_2, \dots, \text{dom}_n\}$. We say that (q', π') is derived in one step from (q, π) , written as $(q, \pi) \xRightarrow{G} (q', \pi')$ if and only if for some $X \in N$ and for some rule $r \in R$ with left part X there exists a subdomain dom_s in π called application area such that

- (i) q' is obtained by substituting the octagonal subpicture x at $\text{dom}_x(q)$ in q with a picture p of same size defined as follows:
 - a) If r is of type (1), then $p = s$
 - b) If r is of type (2), then $p \in L=L(\omega)$.
- (ii) π' is a regional homogeneous partition of $\text{dom}(q)$ into the subdomains $(\pi \setminus \{\text{dom}_{s_i}\}) \cup \text{trans } d_s \pi^*(p)$ where $\text{trans } d_s \pi^*(p)$ denotes the displacement of regional homogeneous partition of p to the position of d_s .

We say that (q', π') is derived from (q, π) in n -steps, written $(q, \pi) \xRightarrow{n}_G (q', \pi')$ if and only if $q = q'$ and $\pi = \pi'$, when $n = 0$ or there is a picture p and a homogeneous partition π'' such that $(q, \pi) \xRightarrow{n-1}_G (p, \pi'')$ and $(p, \pi'') \xRightarrow{*}_G (q', \pi')$. $\xRightarrow{*}_G$ denotes the reflexive transitive closure of $\xRightarrow{G}$.

Definition 3.6:

For $X, Y \in \{R, CF, CS\}$, the (X, Y) Octagonal array language $((X : Y) \text{ OAL})$ generated by the $(X : Y)$ OAG. G is

$$L(G) = \{ T / S \xRightarrow{*}_G T, T \text{ is an Octagonal array} \}.$$

Definition 3.7:

The Octagonal picture language $L(G)$ defined by regional Octagonal tile rewriting grammar (ROTRG) G is a set of $q \in \Sigma_o^*$ such that $\langle s^{|q|}, d(q) \rangle \xRightarrow{*}_G (q, V)$.

V denotes the partition of $\text{dom}(q)$ defined by single pixels. Shortly we can also write $S \xRightarrow{*}_G q$. The family of Octagonal picture languages generated by ROTRG is denoted by $L(\text{ROTRG})$.

Definition 3.8:

A context-free octagonal array grammar $G = \{N, S, R, \Sigma\}$ is in Chomsky-like normal form if and only if every rule in R has the form either $z \rightarrow t$ or $z \rightarrow XY$, where $X, Y, Z \in N$ and $t \in \Sigma$. Here $\circ$ denotes any one of the eight arrowheads, concatenations.

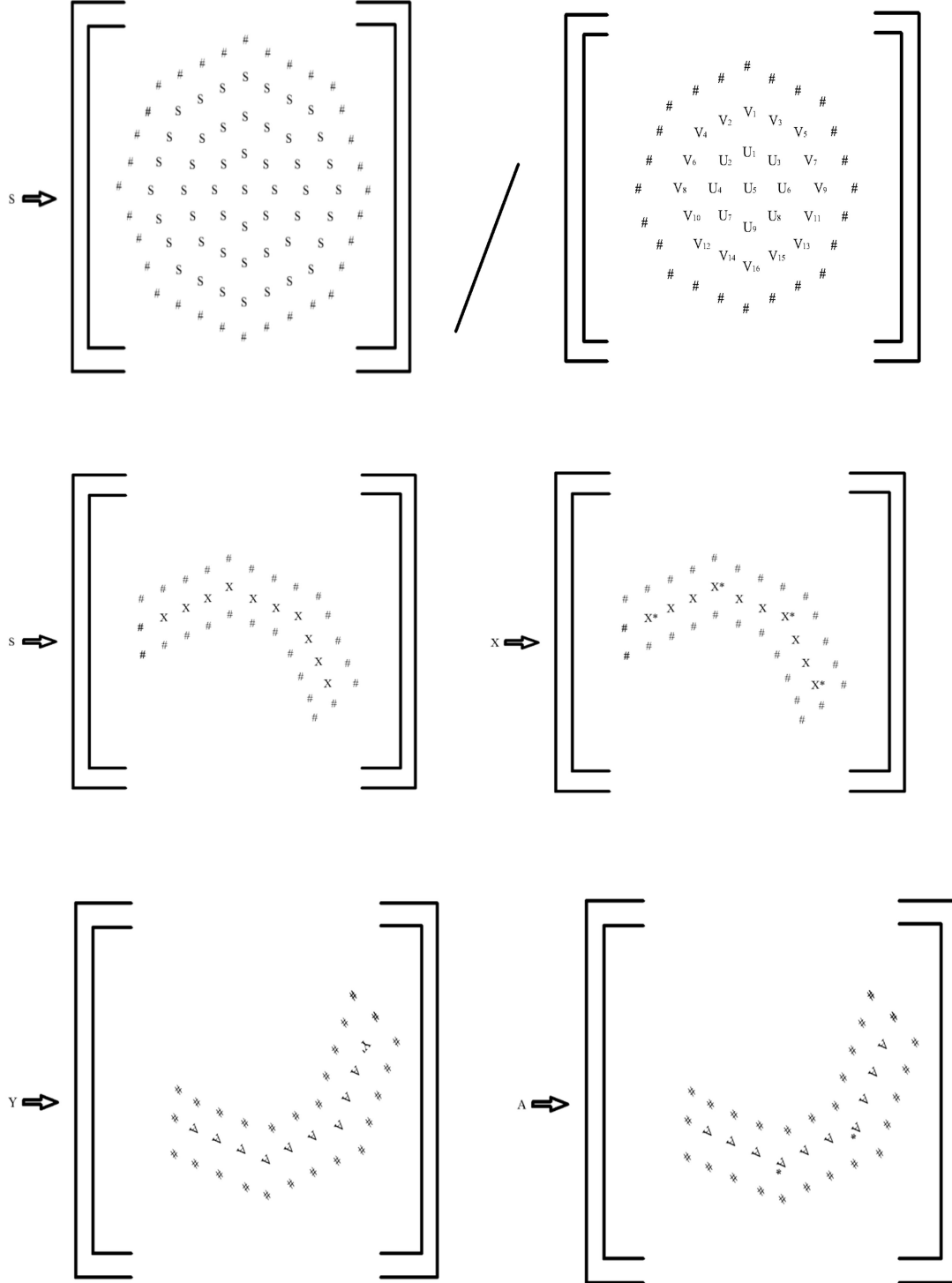
Examples on ROTRG:

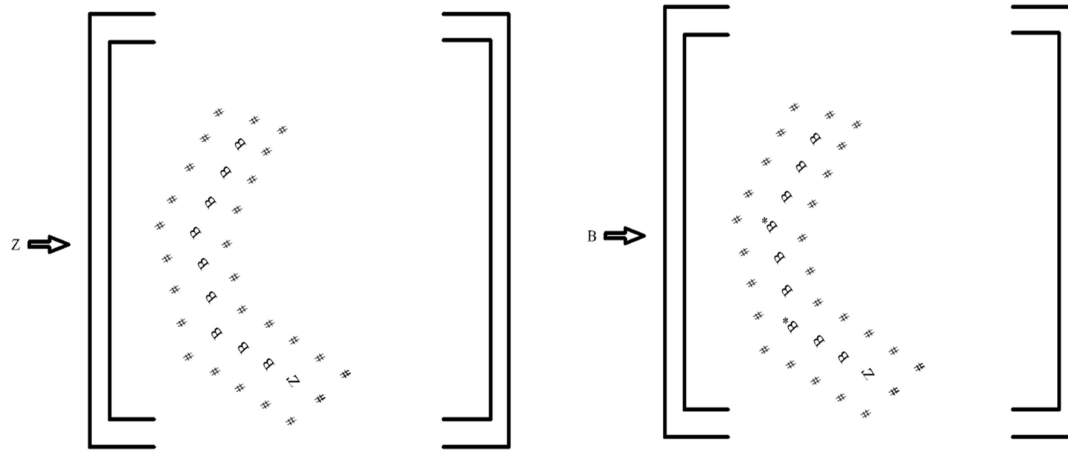
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Eg. 1

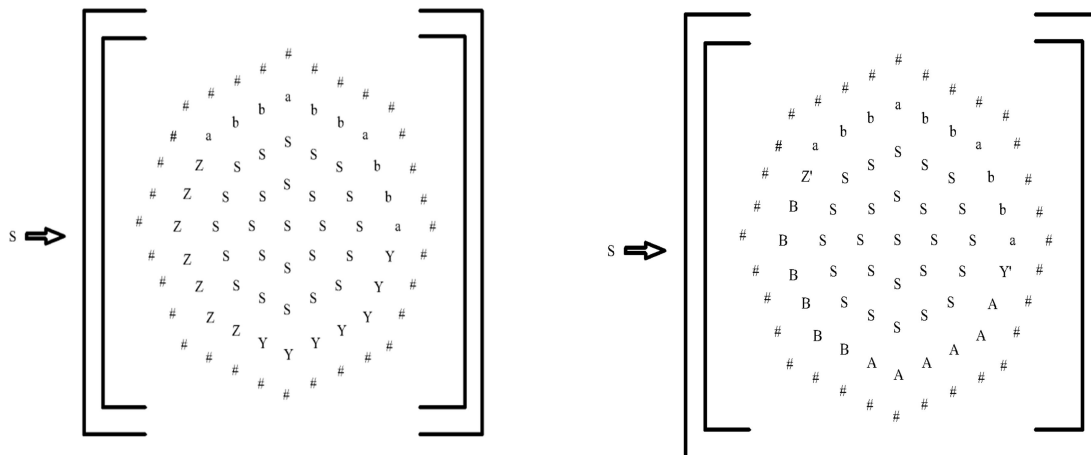
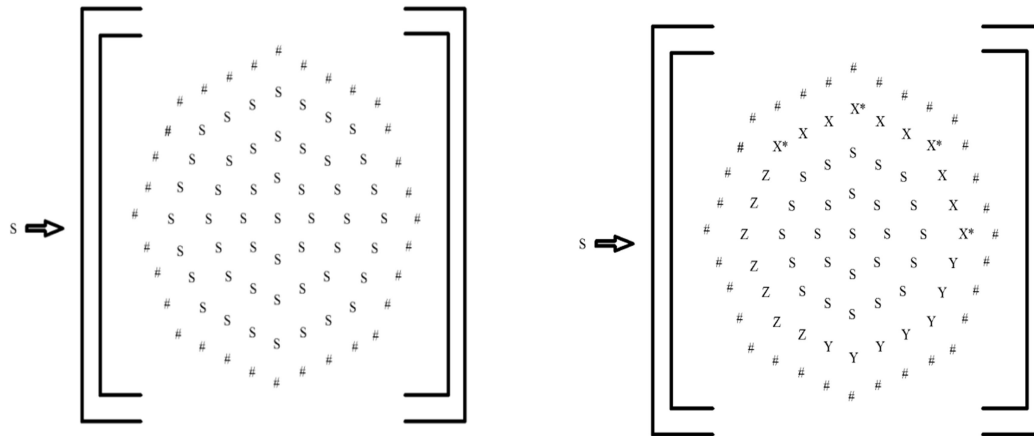
The set of all Octagonal pictures whose elements passes through the vertices are a's and other elements are b's is generated by the ROTRG grammar $G = (\Sigma, N, S, R)$ where $\Sigma = \{a, b\}$, $N = \{X, Y, Z, X^*, Y', Z', A, A^*, B, B^*, V_1, \dots, V_{16}, U_1, \dots, U_9\}$ and R consists of the following rules:





$X^*, A^*, B^*, V_1, V_4, V_5, V_8, V_9, V_{12}, V_{13}, V_{16}, U_1, \dots, U_9 \rightarrow a$

$X, Y', A, Z', B, V_2, V_3, V_6, V_7, V_{10}, V_{11}, V_{14}, V_{15} \rightarrow b$



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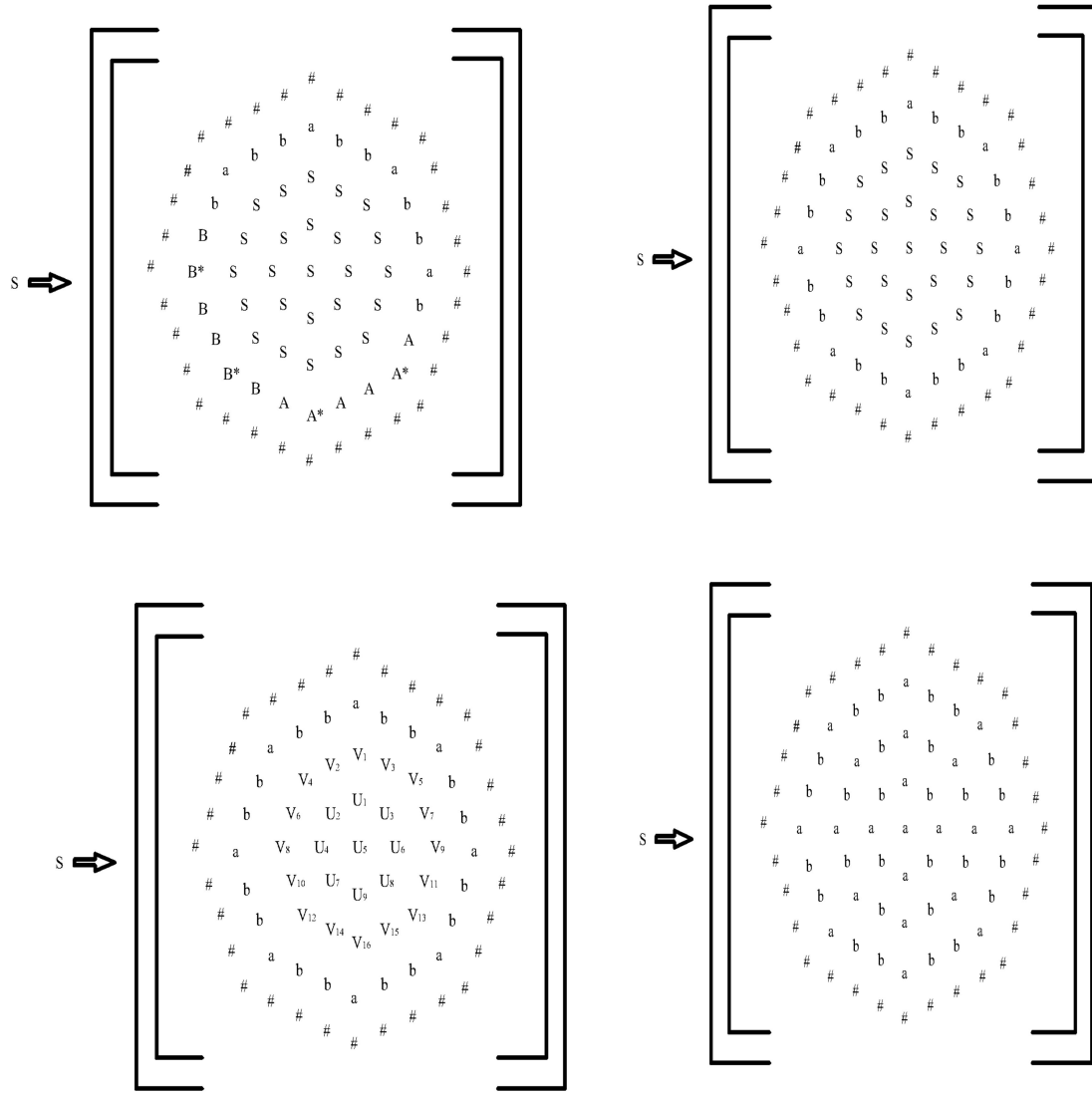
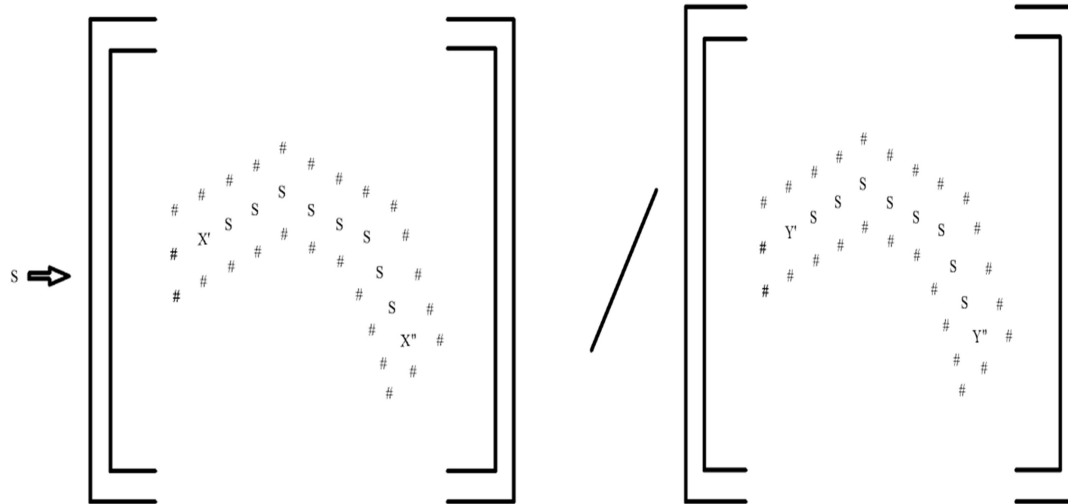


Fig. 2 Consider the language L_1 which consists of palindromic upper arrowheads of thickness one over $\Sigma=\{x,y\}$. A typical element of L_1 is

$$\begin{array}{ccccccc}
 & & & y & & & \\
 x & x & x & & x & x & y \\
 & & & & & & \\
 & & & & & & x \\
 & & & & & & x \\
 & & & & & & x
 \end{array}$$

L can be generated by the Regional Octagonal Tile Rewriting Grammar $G=\{N,S,R, \Sigma\}$, where $\Sigma=\{x,y\}$, $N=\{S, X', X'', Y', Y''\}$ and R having the rules:



$S \rightarrow x / y$
 $X', X'' \rightarrow x$ and $Y', Y'' \rightarrow y$

IV. Comparison Results

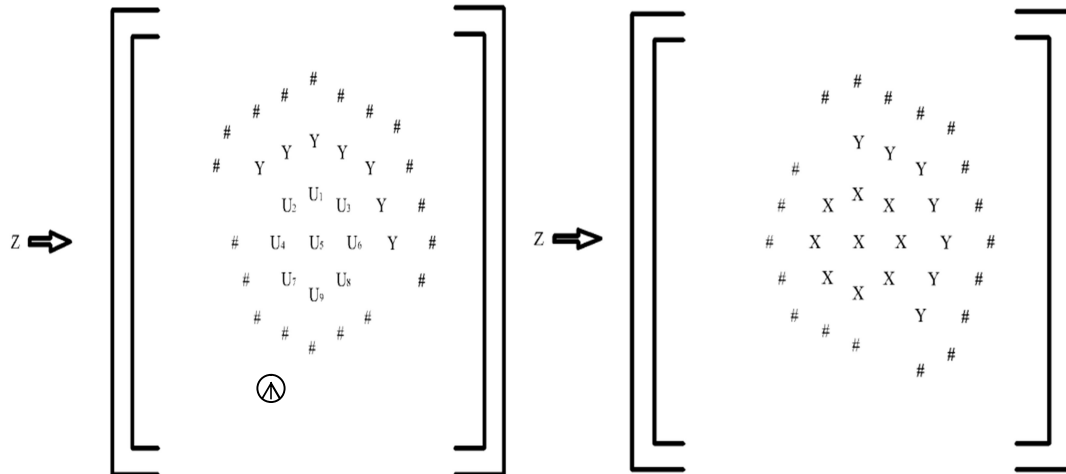
Theorem 1.

The family of languages generated by Context-free Octagonal Array Grammars is strictly included in the family of languages generated by ROTRG. That is $L(\text{CFOAG}) \subset L(\text{ROTRG})$.

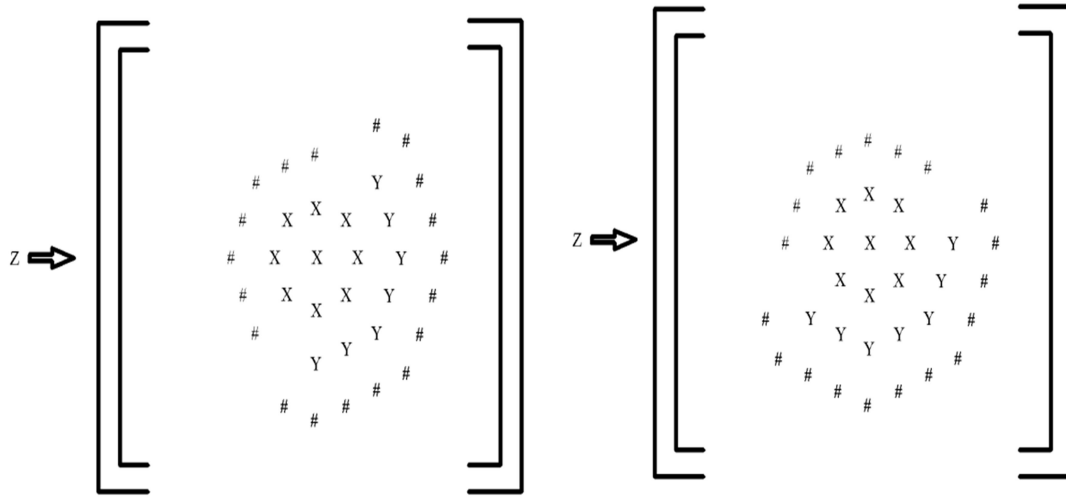
Proof.

Consider a context-free octagonal array grammar (CFOAG), $G = \{N, S, R, \Sigma\}$ in chomsky normal form. Then every rule in R has the form either $z \rightarrow t$ or $z \rightarrow X \quad Y / z \rightarrow X \quad Y / z \rightarrow X \quad Y$ and its dual rules $z \rightarrow X \quad Y / z \rightarrow X \quad Y / z \rightarrow X \quad Y / z \rightarrow X \quad Y$ where $X, Y, Z \in N$ and $t \in \Sigma$.

The rule $z \rightarrow t$ are identical in both context-free octagonal array grammar and regional octagonal tile rewriting grammar models and generate the same kind of languages, that is single terminal symbols. The other rules:



- (i) $z \rightarrow X \quad Y$ of G are equivalent to ROTRG rules of the form
- (ii) $z \rightarrow X \quad Y$ of G are equivalent to ROTRG rules of the form,



- (iii) $z \rightarrow X$ of G are equivalent to ROTRG rules of the form
 (iv) $z \rightarrow X$ of G are equivalent to ROTRG rules of the form

Using CFOAG, we cannot generate the language L (Ex.2), since there is no provision in the rule of the CFOAGE to increase the length of an arrowhead.

It can increase the thickness of the arrowhead but not the length.

Theorem 2.

$L(\text{ROTRG}) \subset L(\text{OTRG})$.

Proof.

Since ROTRG rules are restricted form of OTRG rules, every rule of ROTRG is in OTRG also. Therefore $L(\text{ROTRG}) \subseteq L(\text{OTRG})$. But $L(\text{ROTRG}) \neq L(\text{OTRG})$, as the OTRG language given in the example 3.2 [8] cannot be generated by regional partition. Therefore $L(\text{ROTRG}) \subset L(\text{OTRG})$.

Conclusion

Regional Octagonal Triangular Tile Rewriting Grammars (ROTRG) are the simple type of octagonal tiling based array rewriting models. It has generative capacity more than context-free octagonal array grammars but less than octagonal tiling rewriting grammar. We investigate further interesting properties and practical applications.

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